Integrating Factors, Two Ways

Linear ODEs and inexact differential forms

A calculation-first guide with the geometry underneath

The single idea

An **integrating factor** is a nonzero multiplier μ chosen so that an expression becomes the derivative of something simpler. It does not change the solution curves (away from points where $\mu = 0$); it changes how the differential equation is represented.

1. Linear first-order ODE: make the left side a product derivative

Start with

$$y' + P(x)y = Q(x).$$

We want a function $\mu(x)$ for which

$$\mu y' + \mu P y = (\mu y)' = \mu y' + \mu' y.$$

Matching the coefficients of y requires $\mu' = P\mu$, hence

$$\boxed{\mu(x) = e^{\int P(x)\,\mathrm{d}x}}.$$

Any nonzero constant multiple also works; choosing the constant 1 is simplest.

Reusable algorithm

1. Put the equation in standard form $y' + P(x)y = Q(x)$ (divide by the coefficient of y' first).
2. Compute $\mu = e^{\int P\,\mathrm{d}x}$.
3. Multiply the *entire* equation by μ.
4. Replace the left side by $(\mu y)'$, integrate, and solve for y.

Example: $y' + 2y = e^x$

Here $P = 2$, so $\mu = e^{2x}$. Multiplication gives

$$e^{2x}y' + 2e^{2x}y = e^{3x} \quad \Longrightarrow \quad (e^{2x}y)' = e^{3x}.$$

Therefore $e^{2x}y = \frac{1}{3}e^{3x} + C$, and

$$\boxed{y = \tfrac{1}{3}e^{x} + Ce^{-2x}}.$$

Quick check: $y' + 2y = e^x$. If $y(x_0) = y_0$, substitute after finding the general solution to determine C.

Why the exponential?

The condition $\mu'/\mu = P$ says that the *relative growth rate* of μ must equal the coefficient P. Exponentiating an antiderivative is precisely what solves this condition.

2. $M(x, y)\,\mathrm{d}x + N(x, y)\,\mathrm{d}y = 0$: make a differential form exact

Write

$$\omega = M(x, y)\,\mathrm{d}x + N(x, y)\,\mathrm{d}y.$$

The equation is **exact** when there is a potential $F(x, y)$ such that

$$\mathrm{d}F = F_x\,\mathrm{d}x + F_y\,\mathrm{d}y = M\,\mathrm{d}x + N\,\mathrm{d}y.$$

On a simply connected region with continuous derivatives, this is equivalent to

$$\boxed{M_y = N_x}.$$

Solutions are then the level curves $F(x, y) = C$.

If $M_y \neq N_x$, seek an integrating factor μ such that

$$\mu M\,\mathrm{d}x + \mu N\,\mathrm{d}y = 0$$

is exact. The full condition is

$$\boxed{\frac{\partial(\mu M)}{\partial y} = \frac{\partial(\mu N)}{\partial x}}. \qquad (*)$$

This is generally a PDE for $\mu(x, y)$. Two special tests can reduce it to an ordinary integral.

Fast tests for a one-variable integrating factor

Let $M_y = \partial M/\partial y$ and $N_x = \partial N/\partial x$.

If this depends only on	then try	integrating factor
$\dfrac{M_y - N_x}{N}$	x	$\mu(x) = \exp\left(\int \dfrac{M_y - N_x}{N}\,\mathrm{d}x\right)$
$\dfrac{N_x - M_y}{M}$	y	$\mu(y) = \exp\left(\int \dfrac{N_x - M_y}{M}\,\mathrm{d}y\right)$

These formulas follow directly by substituting $\mu(x)$ or $\mu(y)$ into (*). Always verify exactness afterward.

Worked example: find $\mu(x)$

Consider

$$(2y - x)\,\mathrm{d}x + x\,\mathrm{d}y = 0, \qquad M = 2y - x, \quad N = x.$$

It is not exact because $M_y = 2$ while $N_x = 1$. Now

$$\frac{M_y - N_x}{N} = \frac{2-1}{x} = \frac{1}{x},$$

which depends only on x. Thus, on a region not crossing $x = 0$,

$$\mu(x) = e^{\int (1/x)\,\mathrm{d}x} = e^{\ln|x|} = |x|.$$

On either half-plane $x > 0$ or $x < 0$, a nonzero constant factor is irrelevant, so we may take $\mu = x$. Multiplying gives

$$(2xy - x^2)\,\mathrm{d}x + x^2\,\mathrm{d}y = 0.$$

Now $(2xy - x^2)_y = 2x = (x^2)_x$: exactness is restored.

To find F, integrate $F_x = 2xy - x^2$ with respect to x:

$$F = x^2 y - \frac{x^3}{3} + g(y).$$

Then $F_y = x^2 + g'(y) = N_{\text{new}} = x^2$, so $g' = 0$. The implicit solution is

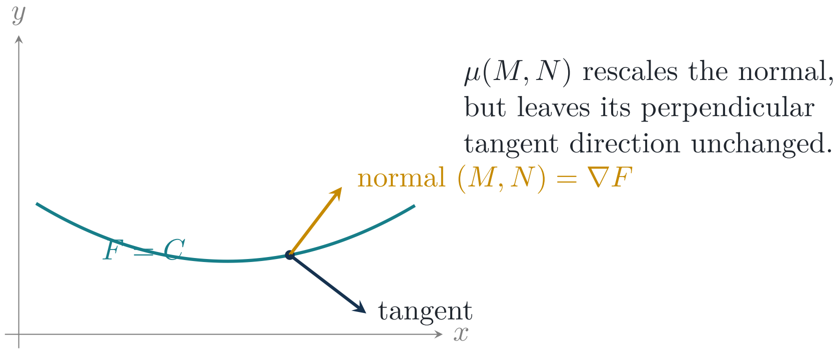
$$\boxed{x^2 y - \frac{x^3}{3} = C} \qquad (x \neq 0).$$

3. Geometry: what multiplication changes

The pair (M, N) can be viewed as a normal vector field. Along a solution curve $(x(t), y(t))$,

$$M\dot{x} + N\dot{y} = 0,$$

so the tangent $(\dot{x}, \dot{y})$ is perpendicular to (M, N). Multiplying by nonzero μ rescales the normal vector but does not rotate it. Thus the allowed tangent direction, and hence the solution curves, stay the same.



Exactness means the rescaled normals fit together globally as the gradient of one height function F. The curves $F = C$ are contour lines on that height surface. In differential-form language, the obstruction to exactness is

$$\mathrm{d}\omega = (N_x - M_y)\,\mathrm{d}x \wedge \mathrm{d}y.$$

An integrating factor cancels this local "curl-like" mismatch so that $\mu\omega = \mathrm{d}F$.

For the linear equation, the same geometry is hiding in the (x, y) plane. Rewrite

$$y' + P(x)y = Q(x) \quad \Longleftrightarrow \quad [P(x)y - Q(x)]\,\mathrm{d}x + \mathrm{d}y = 0.$$

Its integrating factor $\mu(x) = e^{\int P\,\mathrm{d}x}$ makes this form exact, with potential

$$F(x, y) = \mu(x)y - \int \mu(x)Q(x)\,\mathrm{d}x.$$

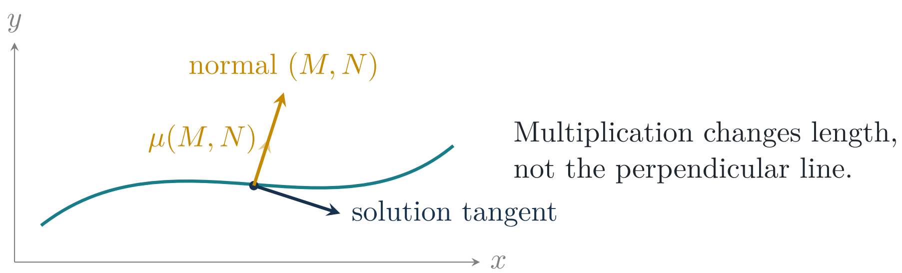
So the "product rule trick" and the $M\,\mathrm{d}x + N\,\mathrm{d}y$ method are not separate magic: the first is a particularly friendly case of the second.

Three pictures to keep in mind

Picture A: a differential form selects directions. At every point, the equation

$$M\,\mathrm{d}x + N\,\mathrm{d}y = 0$$

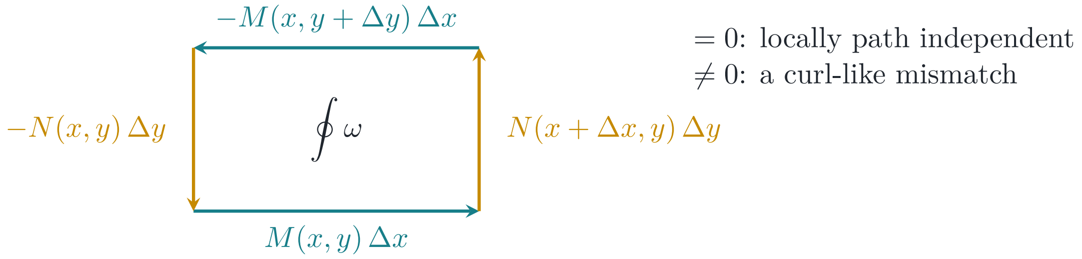
selects displacements $(\mathrm{d}x, \mathrm{d}y)$ perpendicular to (M, N). The field (M, N) is therefore a field of normals, not the direction in which the solution itself travels.



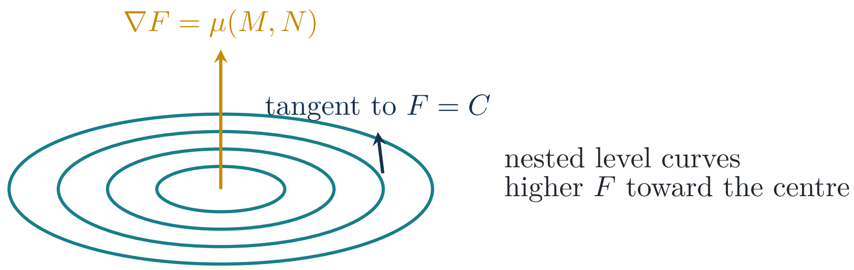
Picture B: exactness is a tiny-loop cancellation test. Traverse a small rectangle of side lengths Δx and Δy. To first order, the circulation of $\omega = M\,\mathrm{d}x + N\,\mathrm{d}y$ around its boundary is

$$\oint_{\partial R} \omega \;\approx\; (N_x - M_y)\,\Delta x\,\Delta y.$$

If $M_y = N_x$, the four edge contributions cancel. If not, the form accumulates a nonzero mismatch around a closed loop, so it cannot be the differential of a single-valued potential.



Picture C: an exact form is a contour map. Imagine a surface $z = F(x, y)$. Its gradient ∇F points uphill, while each solution $F = C$ is a contour line across which the height does not change. When $\mu\omega = \mathrm{d}F$, the rescaled vector field $\mu(M, N)$ is exactly this uphill field.



What the integrating factor really repairs

It does not repair the visible direction field: that already specifies the solution curves. It repairs the *scale from point to point*, choosing arrow lengths so that the rescaled normals can be interpreted consistently as derivatives of one potential function.

4. Decision guide and common traps

Which route should I use?

- If you can write $y' + P(x)y = Q(x)$, use $\mu = e^{\int P\,\mathrm{d}x}$.
- If given $M\,\mathrm{d}x + N\,\mathrm{d}y = 0$, test $M_y = N_x$ first.
- If it is not exact, test the two one-variable ratios. If neither ratio has the required dependence, an elementary one-variable integrating factor is not indicated; solving (*) may require another substitution or a PDE method.

Traps. Divide into standard form before reading off P; multiply every term by μ; keep the signs in the two ratio tests straight; state domain restrictions where μ, M, or N vanish; and verify exactness before integrating for F.

5. Mini practice

1. Solve $y' - \frac{2}{x}y = x^2$ on $x > 0$. *Hint:* $\mu = x^{-2}$.
2. Show that $(y^2 + 2xy)\,\mathrm{d}x + (2xy + x^2)\,\mathrm{d}y = 0$ is exact and find F.
3. For $(y + xy)\,\mathrm{d}x + x\,\mathrm{d}y = 0$, compute $(M_y - N_x)/N$, find $\mu(x)$, and solve implicitly.

Solutions in brief

1. With $P = -2/x$, $\mu = e^{\int -2/x\,\mathrm{d}x} = x^{-2}$ on $x > 0$. Then

$$(x^{-2}y)' = 1, \qquad x^{-2}y = x + C, \qquad \boxed{y = x^3 + Cx^2}.$$

2. Here $M_y = 2y + 2x = N_x$, so the form is already exact. Integrating M with respect to x gives

$$F = xy^2 + x^2y + g(y).$$

Comparison with $F_y = 2xy + x^2 + g'(y) = N$ gives $g' = 0$, hence $\boxed{xy^2 + x^2y = C}$.

3. Here $M = y + xy$, $N = x$, and

$$\frac{M_y - N_x}{N} = \frac{(1 + x) - 1}{x} = 1.$$

Thus $\mu = e^x$. The new form is exact, and integrating $F_y = xe^x$ gives

$$F = xe^x y + h(x).$$

Its x derivative already equals $e^x(y + xy)$, so $h' = 0$ and $\boxed{xe^x y = C}$.

Memory anchor

Linear ODE: *engineer a product rule.* Differential form: *engineer a potential function.* In both cases the multiplier makes the equation a total derivative.

Notation: subscripts denote partial derivatives. All statements are local to regions where the functions are sufficiently smooth and the chosen integrating factor is nonzero.